

Overview

This project proposes the development of a **Multimodal Health AI Model** that enables hospitals to automatically observe, understand, and optimize clinical workflows in real time. The system integrates heterogeneous data sources, including video, audio, wearable inertial measurement unit (IMU) sensors, and electronic health records (EHRs), to reconstruct patient trajectories, provider interactions, and task sequences with precise temporal alignment. Unlike existing clinician-centric AI tools, this research targets hospital-level situational awareness, supporting continuous protocol compliance checking, accurate documentation of care events, and early detection of missing or abnormal clinical actions.

The project advances foundational computing research by addressing key challenges in long-duration multimodal reasoning, multi-view video understanding, and efficient edge system design. By enabling scalable, real-time, and trustworthy AI deployment in complex hospital environments, the proposed system aims to improve patient safety, reduce operational inefficiencies, and lower healthcare costs, while contributing broadly to NSF Future CoRe goals in multimodal AI and edge intelligence.

Keywords: Multimodal Health AI, Hospital Workflow Analysis, Multi-View and Long-Stream Video Reasoning, Edge Computing

Intellectual Merit

This project advances the state of the art in computer and information science by addressing a foundational challenge in future computing: long-duration, real-time multimodal understanding of complex, safety-critical human workflows. The research moves beyond traditional clinician-centric or task-specific medical AI by introducing system-level intelligence capable of reasoning over **multi-actor, multi-view, and multi-modal** hospital environments as they unfold in real time. The intellectual contributions are threefold: (1) We will train hospital-context-aware multimodal AI models that align video, audio, wearable inertial sensor data, and electronic health records into a unified temporal representation. (2) We will design novel algorithms for multi-view and long-stream multimodal reasoning that support continuous monitoring of extended clinical encounters involving multiple caregivers and viewpoints. (3) We will establish a cost-efficient edge system for deploying multimodal AI in hospitals. (4) We will pursue live, multimodal evaluation of LLMs in realistic hospital workflows.

Broader Impacts

The success of this research project will unlock the potential of multimodal AI systems to transform real-world healthcare operations by enabling continuous, efficient understanding of hospital workflows. By providing hospitals with automated, real-time situational awareness, the proposed research has the potential to reduce preventable medical errors, improve protocol adherence, and optimize the coordination of care teams, thereby improving patient safety, clinical outcomes, and overall healthcare efficiency. Moreover, this project will make significant contributions to education and workforce development. It will support the training of undergraduate and graduate students in multimodal AI, edge computing, and healthcare systems. The PIs will incorporate research outcomes into AI- and systems-related curricula. Beyond healthcare, the methodologies developed in this project are applicable to other domains that require long-duration, multi-view, real-time multimodal reasoning.

1 Introduction

Modern hospitals rely heavily on manual documentation, fragmented sensor systems, and retrospective audits to reconstruct what occurred during patient care [3, 47]. These approaches are error-prone, labor-intensive, and often fail to capture the precise timing and sequencing of critical events such as patient transfers, handoffs between providers, and protocol-mandated checks [28]. Deviations from established care pathways, whether due to omissions, delays, or undocumented actions, can increase medical risk, complicate compliance, and lead to unnecessary costs [5]. Consequently, there is a pressing need for intelligent systems that can automatically observe, interpret, and summarize hospital-floor activities as they occur [30].

Fortunately, recent advances in multimodal models have shown promise in partially alleviating this challenge. By jointly leveraging visual, textual, and auditory signals, multimodal large language models (MLLMs) enable a more holistic understanding of complex human activities. For example, prior work has applied multimodal models to surgical video analysis for recognizing procedural phases and detecting anomalies [19, 33], to clinical dialogue understanding and medical report generation [8, 25], and to medical vision-language reasoning for diagnosis and clinical decision support [1, 9]. These studies demonstrate that integrating multiple modalities can improve the accuracy and interpretability of medical AI systems, and provide a foundation for automated perception and reasoning in clinical environments.

However, a significant gap still remains between applications of MLLMs to hospital operations and the capabilities of MLLMs themselves. To close this gap, the following problems need to be addressed: **(1) Unable to understand hospital context:** Most existing MLLMs are trained on generic web-scale data or narrowly defined medical datasets and therefore lack a structured understanding of hospital-specific workflows, roles, and protocols. **(2) Cannot process multiple modalities:** Although recent MLLMs support vision and text, they are not designed to jointly reason over heterogeneous medical data sources, such as video, audio, inertial measurement units (IMUs), and electronic health records (EHRs). **(3) Do not support multiple camera views:** Current models typically operate on single-camera inputs and fail to handle multi-view scenarios involving multiple caregivers wearing body-mounted cameras or interacting across different rooms. **(4) Inefficient processing long stream video:** Most existing multimodal models are designed for short clips or static images and do not scale to long-duration, continuous video streams that characterize real clinical encounters. **(5) Overlook hospital data privacy:** State-of-the-art MLLMs often rely on centralized cloud-based training and inference, which is incompatible with strict hospital data governance and privacy regulations. **(6) Lack of a real and live evaluation:** Most prior work evaluates models using offline benchmarks, simulated environments, or curated datasets, without validating performance in live hospital workflows. This limits the understanding of how these systems behave under real-time constraints, noisy inputs, and complex human interactions.

To address these limitations, this project proposes developing a Multimodal Health AI Model for Tracking Doctors and Patients to Optimize Hospital Operations and Improve Medical Outcomes. Rather than focusing on individual clinician assistance, the proposed research targets the hospital as a whole, aiming to provide an automated, real-time overview of clinical processes. By jointly modeling video, audio, inertial measurement unit (IMU) data, and electronic health records (EHRs), the system will reconstruct patient trajectories, provider interactions, and task sequences with precise temporal alignment. This enables continuous verification of protocol adherence, accurate timestamped documentation of care events, and timely detection of missing or abnormal steps during diagnosis and treatment.

By enabling automated, real-time situational awareness at the hospital level, the proposed re-

search seeks to improve patient safety, reduce operational inefficiencies, and lower healthcare costs. More broadly, this work aligns with the NSF Future CoRe by advancing foundational principles for multimodal AI, edge-cloud systems, and trustworthy computing in complex, real-world environments.

1.1 Project Overview

This project aims to develop a comprehensive multimodal, real-time, and efficient AI system that enables hospitals to automatically observe, understand, and optimize clinical workflows at scale. The overarching objective is to advance future computing foundations that allow long-duration, multi-actor hospital activities to be continuously monitored and reasoned about, thereby improving patient safety, operational efficiency, and medical outcomes while reducing overall healthcare costs.

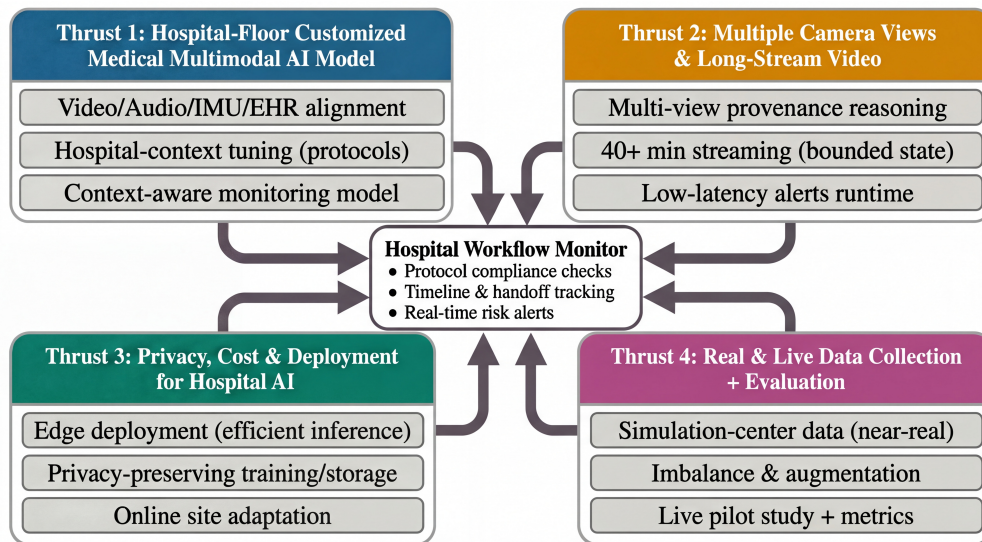


Figure 1: Overview of the proposed four thrusts.

Our investigation is motivated by several key observations that have not been comprehensively addressed in existing literature. (a) Current AI systems in healthcare are predominantly designed for clinician-centric assistance or single-task analysis and do not provide a holistic, hospital-level view of care delivery. (b) Existing multimodal medical AI models are typically trained on short, curated recordings and fail to capture the long-duration, streaming, multi-view videos of real hospital workflows. (c) Most deployed solutions rely on centralized cloud computation and static models, limiting their suitability for privacy-sensitive, latency-critical, and continuously evolving hospital environments. (d) The absence of live or near-realistic evaluation of multimodal large language models in settings that reflect real hospital operations and their impact on care quality and efficiency. As a result, hospitals lack automated mechanisms to verify protocol adherence, reconstruct accurate care timelines, and detect missing or abnormal events in real time.

Moreover, this project is designed to answer the following research question: *Can we develop scalable multimodal AI models and systems that support long-duration, multi-view, real-time understanding of hospital workflows while ensuring data privacy, low latency, and cost efficiency?*

1.2 Intellectual Merit

This project advances the multimodal AI model by addressing the fundamental challenge of long-duration, real-time understanding of complex hospital workflows. By moving beyond clinician-centric assistance toward system-level situational awareness, the proposed research introduces ef-

efficient models, algorithms, and system architectures that enable automated reasoning over multi-actor, multi-view, and multi-modal clinical environments. As illustrated in Figure 1, the intellectual merit of this project is organized around four research thrusts:

- **Thrust 1:** We will develop new multimodal representation learning methods tailored to hospital environments by systematically aligning video, audio, IMU signals, and EHRs.
- **Thrust 2:** We will develop novel algorithms for reasoning over multiple body-worn camera views and continuous video streams.
- **Thrust 3:** We will establish efficient methods for deploying the multimodal model onto edge devices to achieve cost-efficient and real-time inference, and continue updating.
- **Thrust 4:** We will pursue live, multimodal evaluation of LLMs in realistic hospital workflows.

1.3 Team Qualification

The project team brings a complementary and interdisciplinary set of expertise spanning multimodal artificial intelligence, distributed and edge computing, wireless sensing systems, and clinical practice. This combination of strengths is essential for successfully conducting the proposed research, which requires close integration of AI model development, real-time system design, and deployment in authentic hospital environments.

Dr. Omprakash Gnawali, University of Houston (UH): **PI Gnawali** is an expert in sensing systems, AI applications, and Internet of Things with extensive experience designing and deploying large-scale, real-world sensing platforms. His prior work on mobile, wearable, and sensor-rich systems provides a strong foundation for data collection and system integration in complex environments such as hospital floors. His expertise is critical for designing robust multimodal sensing infrastructures, managing real-time data streams from body-worn cameras and IMU devices, and ensuring reliable operation under the dynamic conditions of clinical workflows. He has also worked on a AI-based mortality prediction project that extracted, processed, and utilized the Electronic Health Records data at SUNY Upstate Medical University.

Dr. Chengming Zhang, University of Houston (UH): **PI Zhang** brings deep expertise in machine learning systems, multimodal AI, and efficient model design for resource-constrained platforms, whose works have been published in top conferences, including SC, ICS, PPOPP, CVPR, and NeurIPS. His research focuses on scalable learning algorithms, long-duration video understanding, and edge intelligence, which directly align with the project’s core technical challenges. In [13, 26], he presents an efficient multimodal language model to analyze extensive video data. In [38, 44, 45], He also presented several works on system-algorithm co-design to accelerate AI models.

Dr. Jivan Lamichhane, SUNY Upstate Medical University (Upstate): **PI Lamichhane** provides essential clinical domain expertise and direct access to real-world inpatient environments. As a practicing clinician-scientist, he brings a deep understanding of hospital operations, patient-safety requirements, and clinical decision-making under real workflow constraints.

2 Related Work

Health Multimodal Large Language Model: Recent advances in multimodal large language models (MLLMs) have significantly expanded the scope of medical video analysis by enabling joint reasoning over visual, temporal, and linguistic information. Early work adapts instruction-tuned vision-language models to medical videos, enabling video-grounded dialogue and open-ended clinical reasoning over surgical and endoscopic recordings [21, 34], with subsequent studies emphasizing multi-grained temporal modeling to capture long-term procedural structure for full-procedure understanding [35]. In parallel, foundation video-language pretraining approaches leverage large-scale

unlabeled medical videos to learn transferable representations, enabling zero-shot recognition, retrieval, and cross-procedure generalization via contrastive or hierarchical alignment [42, 43], with recent extensions incorporating LLM-generated procedural knowledge and explicit temporal constraints [41], and further expanding to other medical video modalities such as ultrasound and video-level VLM architectures [12, 23].

Despite these advances, existing healthcare MLLMs remain largely assistant-centric and episodic, operating on short, curated inputs rather than continuous multimodal streams. Most approaches consider limited modality combinations and do not jointly model video, audio, inertial sensing, and longitudinal EHR data, nor do they capture hospital-level context such as workflow dynamics, staff movement, or evolving patient state. These limitations restrict the applicability of current systems to real-world hospital environments and motivate the need for new multimodal representations tailored to continuous clinical settings.

Long and Stream MLLMs: Understanding long-horizon and streaming visual data is an active area of research in both computer vision and multimodal learning. Prior work on surgical video analysis has employed temporal convolutional networks and transformers for phase recognition and procedural understanding [6, 24], while recent MLLM-based approaches align video with language supervision to support fine-grained reasoning [21, 34, 35, 41–43]. Large-scale datasets such as Ego-Exo4D further demonstrate the importance of modeling first- and third-person perspectives in complex activities [10].

More recently, streaming and long-video MLLMs have been proposed to handle extended video sequences and online inference by leveraging hierarchical temporal representations and memory mechanisms [4, 29]. However, existing approaches are typically limited to single-view or fixed-camera settings and do not scale to multi-view hospital floors with asynchronous, continuous streams. Moreover, integration with non-visual modalities and patient-specific context from EHRs during streaming inference remains largely unexplored, limiting their utility for real-time clinical workflows.

3 Research Plan

3.1 Thrust 1: Advancing Hospital-Floor Customized Medical Multimodal AI Model

The objective of this thrust is to develop a new class of **hospital-floor-customized multimodal AI models** that can perceive, interpret, and reason about complex clinical workflows across time, space, and multiple human actors. Most existing medical AI systems are limited to (1) narrowly scoped problems such as medical image classification, clinical note generation, or diagnostic recommendation [8, 9]. As a result, they cannot answer operationally critical questions such as: *Has a required safety check been performed? Was a patient transfer properly documented? Did the correct sequence of clinical actions occur?* (2) lack of coherent alignment across multiple modalities. While recent models can process vision and text, they are not designed to jointly reason over video, audio, inertial measurement unit (IMU) data, and electronic health record (EHR) in a unified manner.

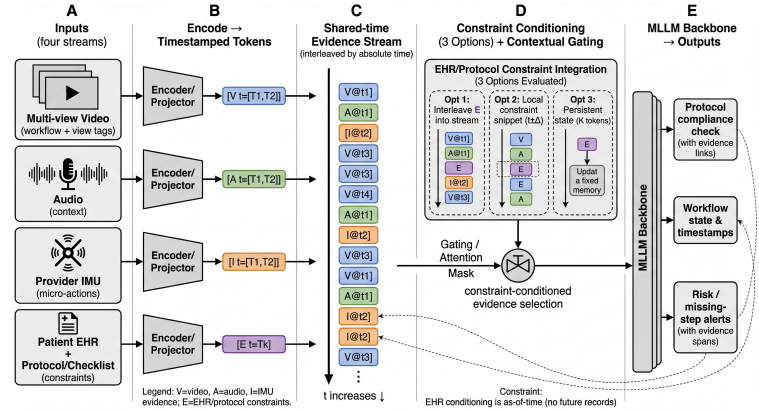


Figure 2: Task 1.1: align video, audio, IMU, and EHR onto a single absolute timeline. Each modality is encoded as timestamped tokens interleaved along the time stream, while EHR context provides gating constraints on cross-modal communication.

3.1.1 Task 1.1: Fine-tune model to understand the hospital context.

Current MLLMs are strong on video and audio, but they are not designed for hospital workflow monitoring. Hospital context is not limited to video/audio: it also includes wearable IMU signals and EHR records that describe the patient’s current condition and the rules to be followed. Recent advanced models, such as Qwen-Omni [39], demonstrate impressive native processing of audio and vision; they are not inherently designed for the rigid constraints of hospital workflow monitoring. Clinical context extends beyond audiovisual cues to include wearable IMU signals, which capture granular motion, and EHR, which define patient status and active protocols. The inability of existing models to co-register these heterogeneous data sources creates a fundamental *data disconnect*: continuous sensor evidence (*what occurred*) remains isolated from time-anchored patient constraints (*what should occur*). Without this alignment, models cannot reliably link actions to patient-specific restrictions, making protocol-aware monitoring difficult to operationalize.

To bridge this gap, we will develop a *timestamp-aligned four-stream interface*. As shown in Figure2. By unifying video, audio, IMU, and EHR data into a single cohesive timeline, our proposed architecture aims to shift the paradigm from generic anomaly detection to protocol-grounded monitoring.

First, to incorporate high-frequency IMU data without saturating the context window, we will design a lightweight projector (e.g., a TCN or 1D-CNN) that maps raw IMU signals to compact *IMU Evidence Tokens*. Each token will represent a fixed temporal window, explicitly anchored to the global timeline via device timestamps. Crucially, we plan to implement offset calibration against audio/visual clocks to ensure cross-modal synchronization, enabling the system to preserve fine-grained motion cues while adhering to tight latency budgets.

Second, we will treat EHR data not as a static context, but as a dynamic stream of constraint tokens. Structured records (e.g., vitals, isolation status) will be encoded as timestamped key-value-unit tokens, while unstructured notes will be processed into clinical-text tokens. To optimize the fusion of sparse EHR updates with dense sensor streams (aligned via Qwen-Omni encoders), we intend to evaluate three integration strategies: (1) *strict chronological interleaving*, (2) *update-triggered insertion* (injecting constraints only near relevant sensor data), and (3) a *persistent constraint state*. The optimal strategy will be selected based on its balance of temporal fidelity and robustness to delayed updates.

Finally, we will fine-tune a *constraint-gated parameter-efficient fine-tuning (PEFT) bridge* a small gating-and-connector module inserted between multimodal evidence tokens and the LLM, that uses time-active EHR constraints (e.g., isolation) to gate sensor-to-text fusion, so the same action may be flagged as a violation under isolation but not otherwise, reducing constraint-irrelevant false alarms. The same action can have different implications depending on the patient’s state. For example, it may be a violation when the patient is in isolation, but acceptable when they are not; the model will use the EHR state at that time to decide. We will freeze the backbone and apply LoRA-style low-rank updates only to the modality connectors and gating module [16, 17]. We will further add counterfactual pairs by swapping/editing active constraints (with consistent timestamps) to teach that identical evidence can be compliant or non-compliant under different clinical contexts [2].

3.2 Thrust 2: Advancing Support for Multiple Camera Views and Long Stream Video

Building on Thrust 1’s timestamp-aligned four-stream interface (time-stamped evidence tokens + time-anchored EHR constraint tokens with gated cross-modal conditioning), Thrust 2 targets real deployments with concurrent body-cams/room views over long encounters (30–60+ minutes). The challenge is to preserve provenance (who did what) under scalable, latency-bounded streaming for proactive alerts. Most MLLMs remain turn-based [40], and existing streaming pipelines often rely

on chunked ingest-then-generate loops that introduce latency and boundary artifacts [4, 29]. We address this with a unified multi-view, long-stream framework with bounded token/state growth (Figure 3): Task 2.1 develops provenance-preserving multi-view tokens with explicit source/role tags and fixed-budget summaries, and Task 2.2 designs a clocked multi-track engine with bounded-rate state updates that decouples perception from alert generation.

3.2.1 Task 2.1: Support multiple camera inputs.

In a mobile camera network, the same clinician may be observed in multiple streams (their own body-cam, other body-cams, and the room view). We therefore treat ‘where’ as camera/time association (which streams contain evidence about the actor at a given tick), rather than fragile biometric re-identification. In this task, we will focus on a provenance-preserving multi-view representation that remains computationally feasible under concurrent high-resolution streams (e.g., Doctor/Nurse body-cams and a room view). Here, *provenance-preserving* means the representation maintains traceable *who-did-what* attribution even under occlusions and viewpoint changes, without relying on brittle visual identity recognition. Our guiding principle is that each view remains an *independent* token sequence with explicit tags, and cross-view integration occurs through attention rather than irreversible feature pooling. Figure 4 overviews Task 2.1: each camera stream is encoded into a tagged token sequence (view/role/time), summarized into a fixed token budget, and integrated via cross-view attention to produce an attribution-preserving multi-view representation.

First, we will design source-tagged, time-aligned multi-view token packing. A naive approach that spatially stitches views into a fixed “Image Grid” wastes capacity under **modal-density mismatch** [20]: hand-centric views can exhibit high semantic density while room/wall views often contain near-zero useful signal, so fixed grids downsample critical micro-cues while wasting compute on redundant backgrounds. Moreover, late-fusion pooling/averaging across views can blur provenance and entangle actor identity [11], which is unacceptable for compliance monitoring where attribution is as critical as the action itself. Therefore, we will encode each view i as a separate token sequence $V^{(i)}$ and prepend a learned source/view tag $E_{\text{src}}^{(i)}$: $\tilde{V}^{(i)} = [E_{\text{src}}^{(i)}; V^{(i)}]$. We will pack view sequences by a shared timeline (clocked windows) so that attention can align evidence across views while retaining traceable source identity. To handle heterogeneous fps and dropped frames, we will incorporate missingness indicators and view-time offset embeddings to stabilize cross-view alignment under real deployments.

Second, we will ground attribution using role/identity conditioning from deployment metadata. To avoid fragile visual identity recognition, we will condition each wearable stream on the wearer’s professional role using **Identity-Preserving Role Embeddings** $E_{\text{role}} \in \{E_{\text{Dr}}, E_{\text{Nurse}}, E_{\text{Tech}}\}$ retrieved from **device pairing logs**. We inject the role prior as a Role Token: $H_{\text{stream}}^{(i)} = [E_{\text{role}}^{(i)}; \tilde{V}^{(i)}]$. If multiple actors share the same role, we additionally prepend a device/actor ID token $E_{\text{id}}^{(i)}$. We will quantify robustness to pairing noise (e.g., temporary handoffs) and implement a fallback based on device-ID conditioning plus periodic re-pairing checks when role logs are missing or unreliable.

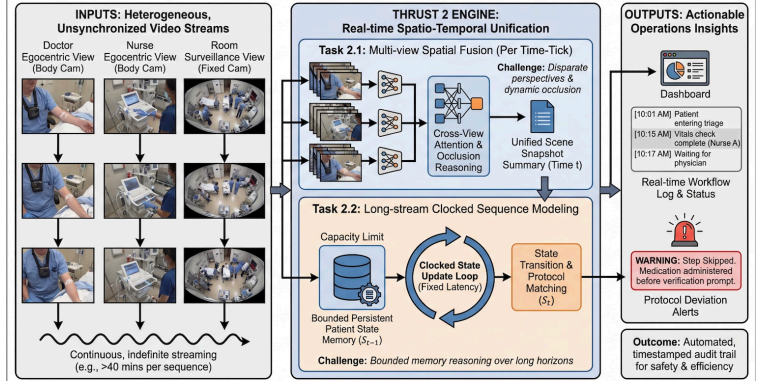


Figure 3: Thrust 2’s overview.

Then, we will develop fixed-budget per-view summary tokens and expose a variable-rate evidence interface. To prevent token explosion as the number of views grows, we will compress each view into a fixed number K of learnable **Summary (Meta) Tokens** using a fixed-query summarization module inspired by TokenLearner/Perceiver-style tokenization [18, 32].

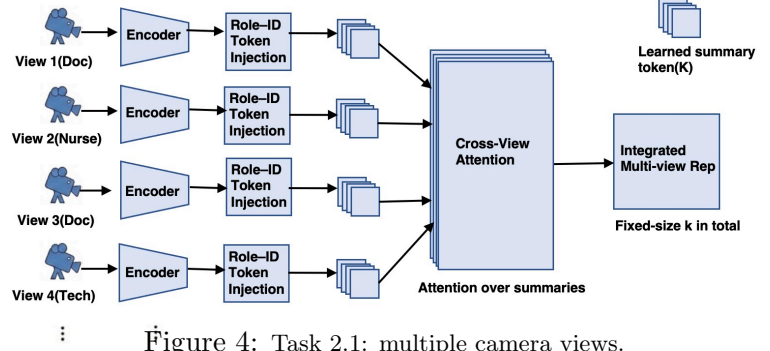


Figure 4: Task 2.1: multiple camera views.

By default, cross-view integration will operate primarily over per-view summary tokens (and optionally sparse frame tokens), yielding a fixed-size multi-view representation per window. We will additionally expose an *interface* that can accept denser tokens for selected views when higher-fidelity evidence is required; the policy for selecting views and allocating tokens is implemented and evaluated in Task 2.2.

Third, we will conduct a systematic study of the accuracy–efficiency trade-off of the proposed provenance-preserving multi-view representation and summarize practical deployment guidelines. For simplicity, we will first fix a default packing/windowing scheme and a fixed summarization budget K , and evaluate (i) attribution accuracy and (ii) provenance stability (*who-did-what* consistency) under occlusions and viewpoint changes. Subsequently, we will incrementally increase the number of concurrent views and the level of pairing noise (e.g., temporary handoffs) to identify a threshold where attribution degrades significantly. Meanwhile, we will track token/compute cost per minute and summarize the scaling trend under a fixed-budget setting. The observed trends will guide how to choose K and when to allocate denser evidence to selected views (handled in Task 2.2). We will also report an ablation matrix and actionable guidelines that connect design choices (source/role tags, offset/missingness embeddings, and summary tokens) to the attribution–compute trade-off.

3.2.2 Task 2.2: Support for long-stream video.

Next, we will focus on a *clocked multi-track streaming engine* that supports 40+ minute encounters without unbounded context growth, while enabling low-latency alerts during continuous ingestion. A *clocked* engine updates a bounded monitoring state at fixed ticks, avoiding explicit chunk boundaries that introduce boundary artifacts. We further decouple monitoring from alert generation so that generated text cannot contaminate the monitoring state.

Specifically, we will pursue the following steps. First, we will implement clocked state updates with fixed-size Meta-State. We discretize time into ticks of length δ seconds. At each tick, we compress each modality track into a fixed-size summary and update fixed-size Meta-State Tokens $S_{t,1:K}^{(s)}$ using tick observations $\hat{X}_t^{(s)}$ and a time-delta embedding Δt :

$$S_{t,1:K}^{(s)} = \text{Update}_{\phi}\left(S_{t-1,1:K}^{(s)}, \hat{X}_t^{(s)}, \Delta t\right). \quad (1)$$

We will reuse Task 2.1 per-view summaries as video/audio inputs when available. Typed EHR events will be encoded as sparse event tokens (event type, value, timestamp) and injected into the corresponding tick update. For IMU, we will fuse body-part channels into a compact per-actor micro-action state within the tick (person-level fusion) and inject this state into the update. Only Meta-State Tokens persist across ticks; raw window tokens are discarded after the update, yielding bounded context over long encounters. We will benchmark alternative update operators (e.g., gated recurrent-style vs. cross-attention updates) and quantify long-horizon drift versus compute.

Second, we will build parallel perception–alert streaming with one-way attention. We will implement two parallel streams: a *perception stream* that updates Meta-State and an *alert stream* that decodes notifications. We decouple the streams via independent positional clocks and a one-way causal attention mask: perception attends only to past perception/state, while alerts attend to all observed perception/state plus past alerts, with no information flow from generated alerts back into perception. This one-way design prevents alert text from contaminating the monitoring state and enables overlapped ingestion (prefill) and alert decoding (decode) without explicit chunk boundaries, reducing notification latency.

Third, we will enforce budgeted scheduling under a fixed real-time constraint and quantify robustness. We enforce a fixed global per-window token budget B aggregated across views and modalities, where B is chosen by profiling the target model/hardware to satisfy a real-time deadline (bounded prefill/state-update latency) and memory constraints. The engine will (a) select a small set of high-salience views for temporary dense refresh using lightweight triggers (e.g., motion energy or audio VAD) and (b) compensate by reducing token rates on inactive views to keep the global budget fixed. We will evaluate robustness to boundary artifacts and long-horizon drift under irregular sampling, dropped frames, and bursty activity, and we will quantify the trade-off between clinically-critical workflow-step miss rate and end-to-end latency.

We abstract the above design into a practical methodology for long encounters under fixed real-time constraints. Specifically, we first profile the target model/hardware to choose a feasible operating point (token budget B , tick size δ , and Meta-State size K). We then implement tick-based bounded state updates with per-track summaries (reusing Task 2.1 view summaries when available) and aligned injections of sparse EHR/IMU signals. Finally, we build a parallel perception–alert runtime with one-way attention and budgeted scheduling to reduce boundary artifacts while meeting real-time deadlines. We will evaluate alert latency (incl. tail), state/token growth per minute, and long-horizon workflow recall plus provenance stability, and summarize guidelines mapping $(\delta, K, \text{update operator, scheduling policy})$ to latency–accuracy trade-offs.

3.3 Thrust 3: Advancing the System for Hospital Data Privacy and Cost.

The objective of this thrust is to develop a **privacy-preserving, cost-efficient, and real-time** system architecture for deploying multimodal hospital intelligence in real-world clinical environments. This thrust addresses a critical barrier to practical adoption: while large multimodal models show strong potential for clinical reasoning, their reliance on centralized cloud computation conflicts with hospital requirements for data privacy, low latency, regulatory compliance, and operational cost control.

3.3.1 Task 3.1: Edge-based deployment of multimodal AI models.

In this task, we will investigate methods for porting large multimodal AI models to resource-constrained edge devices (e.g., NVIDIA Jetson) for deployment directly within hospital infrastructure. First, to reduce the number of input tokens, we will design an event-centric method to merge redundant input modalities. Video, audio, IMU, and EHR signals are modeled as a dynamic sequence of clinically meaningful state transitions rather than as uniform streams. Second, to further reduce the computational footprint, we will apply structured layer pruning and parameter compression to the multimodal model. Redundant layers and attention heads will be identified and removed using sensitivity analysis and task-aware pruning criteria. In addition, we will perform weight quantization and key-value (KV) cache quantization, significantly reducing memory usage and improving inference speed. Third, we will develop a layer-by-layer computation scheduling mechanism that executes the model incrementally in small blocks rather than loading the full model into memory.

3.3.2 Task 3.2: Online and continuous fine-tuning on edge devices.

Hospital environments are non-stationary: workflows evolve, staff behaviors change, new protocols are introduced, and patient populations shift over time. Static models trained offline will inevitably degrade in performance unless they can adapt continuously. Thus, we will develop memory-efficient and computation-efficient online learning algorithms that enable the multimodal model to be updated directly on edge devices.

Specifically, we will propose an **importance-aware** multimodal LLM low-rank fine-tuning framework to determine low-rank configurations under the fixed computation and memory budget. Parameter-efficient fine-tuning (PEFT) methods, particularly Low-Rank Adaptation (LoRA), have become a popular solution. Existing PEFT methods typically adopt a uniform rank configuration across all layers and modules, assuming equal importance among them. This uniform treatment leads to inefficient parameter usage. To address these issues, we will let the unimportant layers/modules freeze or assign them a lower rank. We will quantify the importance of individual weight matrices across the many layers and modalities of an MLLM via effective rank, which measures the intrinsic dimensionality of a matrix by computing the entropy of its normalized singular value distribution. Formally, given a matrix $\mathbf{M}$, its effective rank is: $r_{\mathbf{M}} = \exp(-\sum_k p_k \log p_k)$, $p_k = \frac{\sigma_k}{\sum_j \sigma_j}$ where σ_k are the singular values of $\mathbf{M}$. A higher effective rank indicates greater expressiveness and importance of the corresponding module. Finally, we will implement a privacy-preserving adaptation pipeline, in which all online learning occurs locally on the edge device. This design ensures that sensitive patient data never leave the hospital while still enabling collective learning across institutions.

3.4 Thrust 4: Real and Live Data Collection for Model Training and Evaluation.

A critical gap in current research is the absence of live or near-realistic evaluation of multimodal large language models (LLMs) in settings that reflect real hospital operations and their impact on care quality and efficiency. Nearly all prior work relies on static datasets or narrowly scoped modalities, limiting external validity and practical relevance. This project is among the first to pursue live, multimodal evaluation of LLMs in realistic hospital workflows.

Because such evaluations raise methodological, logistical, and regulatory challenges, a core thrust of this work is the systematic development of protocols, data collection methodologies, institutional collaborations, and IRB-compliant procedures needed to execute live studies successfully.

3.4.1 Task 4.1: Addressing lack of near-realistic multi-modal data

Our objective is to model team-based, multi-vantage hospital events that unfold over time and across modalities. We conduct controlled yet realistic simulations at the SUNY Upstate Simulation Center, a clinical training facility. Standardized patients are recruited to enact clinically authentic scenarios, while teams of physicians and nurses deliver care under typical hospital workflows. Scenarios are designed to include both optimal and sub-optimal decisions, capturing workflow breakdowns and coordination failures that are rare in static datasets but central to real-world safety and efficiency analysis. During the simulations, providers wear professionally mounted GoPro cameras to capture first-person video, along with inertial measurement unit (IMU) sensors on their forearms and upper arms to record fine-grained motion. This multimodal setup enables synchronized analysis of visual context, team interaction, and physical actions. Following IRB approval, a pilot data collection involving multiple scenarios was successfully done in January 2026. (Figure 5)

3.4.2 Task 4.2: Addressing imbalanced datasets in realistic settings

While live data collection is essential, sub-optimal care events are inherently sparse in operational U.S. hospitals, leading to significant class imbalance. We address this challenge through two com-



(a) Nurse's view while walking in (b) Resident doctor's view while the nurse checking the pulse (c) Attending physician's view while the nurse is readying the stethoscope (d) Attending physician's view, resident after the main pulse is taken (e) Nurse's view after the main provider interviewing the patient

Figure 5: Thumbnails from multiple video vantages and over time while a team of providers are taking care of a standardized patient at the SUNY Upstate simulation center in January 2026.

plementary strategies:

1) Annotated medical television program data. We leverage medical television programs (Grey's Anatomy during the pilot) as a source of richly contextualized clinical interactions. As a pilot, a University Hospital board certified medical faculty reviewed randomly selected episodes of Gray's Anatomy available on Netflix and identified eleven distinct sub-optimal practices spanning multiple severity levels. During the project, we will scale this process to generate several hundred expert-annotated clips capturing diverse workflow failures and communication breakdowns.

2) Enacting scripted errors at the simulation center: We systematically design simulation scenarios at the SUNY Upstate Simulation Center that incorporate common, high-impact clinical mistakes, informed by real hospital experience. Examples from the January simulation campaign include 1) delayed fall-risk mitigation in syncope, 2) delayed rapid response in anaphylaxis, and 3) incomplete hand hygiene during routine care. These controlled errors can be reliably reproduced and labeled, producing high-value training data. Over the project duration, we plan six simulation campaigns, each targeting different categories of anomalies.

Collected data will be further enhanced through data augmentation techniques, enabling robust model training despite inherent imbalance [7, 15]. We address data scarcity via a Generative Counterfactual Engine. First, we utilize diffusion-based video inpainting/editing to synthetically inject safety hazards into compliant video, manufacturing rare, automatically labeled "hard negative" samples that were never physically recorded [22, 46]. Second, we apply biomechanical simulation to IMU streams, warping trajectories to model diverse provider physiologies [15]. Finally, using sliding window sampling and modality dropout / missing-modality robust training, we amplify the 120 seed sessions into over 14,000 distinct context windows [27, 31, 36, 37]. This strategy transforms focused observational data into a dense, high-entropy training corpus, ensuring robust generalization for safety-critical fine-tuning.

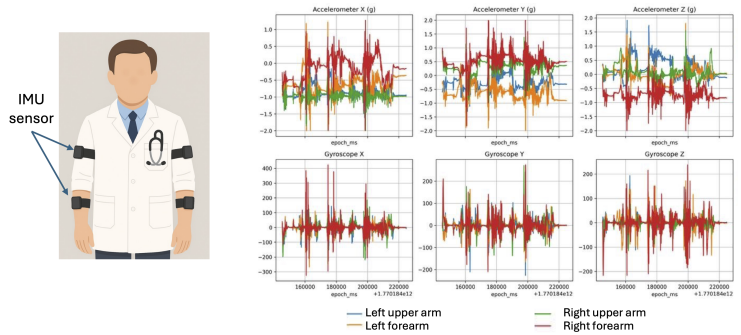


Figure 6: Four wearable IMU sensors are strapped on the upper arm and forearm of a physician. The data represents a provider trying to extend their arm to prevent the patient from falling and bumping the patient in to the recliner chair.

3.4.3 Task 4.3: Addressing lack of real-world evaluation

We will conduct live data collection on a designated hospital floor at SUNY Upstate Medical University. Providers will wear GoPro cameras and IMU sensors during routine care, and corresponding

EHR documentation will be obtained to provide structured clinical context.

Given privacy constraints and variable provider opt-out rates, data collection will be limited to clearly marked zones and will proceed only with explicit consent from the patient, all participating providers for the patient and any nearby patients who may be captured in the videos. Prior experience and ongoing outreach by the SUNY Upstate PI team suggest that sustained engagement will enable collection of 100–150 fully consented care sessions over the project period. A preliminary IRB for this live phase is currently under consultation with SUNY Upstate medical leadership.

Integration with EHR data (structured components and clinical freeform notes) will enable alignment of sensor streams (Video, IMU) with clinically meaningful timestamps (e.g., procedure initiation, medication administration). While certain anomalies may not be explicitly (intentionally or not intentionally) documented in EHRs, these anchors provide essential grounding for multimodal analysis and evaluation. Prior work by PI Gnawali and PI Lamichhane on EHR-based patient mortality prediction at SUNY Upstate provides a strong foundation for this integration.

To ensure broad coverage and generalization, the full dataset will span multiple departments, patient demographics, and care pathways. During the pilot, before scaling, we will validate the end-to-end data pipeline: sensor deployment, synchronization, annotation, approvals, logistics, and quality control. Data collection will preserve ecological validity, with no artificial time constraints imposed in patient care.

4 Preliminary Results

Streaming Performance of Our Model on Medical Data. We evaluate our unified multi-view streaming system on long-form clinical videos from MedVidQACL [14], which contains expert-annotated instructional steps and supports visual answer localization. We focus on full-length diagnostic sequences (typically ~ 40 minutes) to stress-test continuous monitoring and evidence retention beyond short-clip settings. We present three representative case studies (Figs. 7–9) to illustrate how different design choices affect clinical workflow understanding under the same overall context budget. **Result 1** isolates the effect of *cross-stream context sharing* (siloes vs shared context) on appropriateness judgment. **Result 2** isolates the effect of *multi-view evidence* (single-view vs multi-view) on *who-did-what* attribution. **Result 3** isolates the effect of *query-guided checklist prompting* on long-horizon error localization and evidence grounding.

Results 1: Unified Model Improves Appropriateness Judgment. We first study whether a *single unified model* with shared context can make more reliable appropriateness judgments than *siloes per-view/per-role models* without context sharing. Given the same clinical question (e.g., whether the observed action is appropriate under the patient’s vital state), siloes models often lack key evidence that resides in another stream (e.g., monitor readings or the initiating actor), leading to under-specified or inconsistent judgments (Fig. 7).

In contrast, our unified model integrates complementary cues (e.g., severe bradycardia from the monitor and concurrent chest compressions), enabling protocol-consistent assessment with explicit supporting evidence.

Results 2: Multi-view Resolves Actor–Action Ambiguity. Next, we evaluate single-view versus multi-view perception for attributing *who did what*.

As shown in Fig. 8, Single-view inputs frequently yield partial observations (occlusions or missing context), making it difficult to disambiguate the responsible actor and the precise intervention. Our multi-view representation keeps

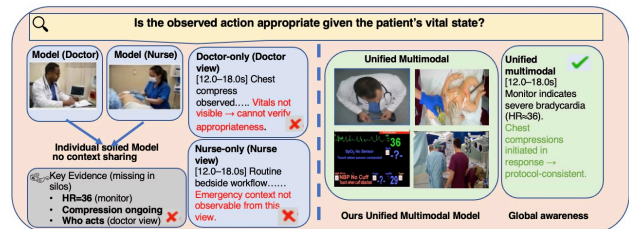


Figure 7: Unified Multimodal Context Sharing for Appropriateness Judgment

each view as a source-tagged stream and aggregates evidence via cross-view reasoning,

allowing the unified model to correctly attribute interventions (e.g., identifying chest compressions performed by the doctor with the nurse assisting) when any one view alone is ambiguous.

Results 3: Query-Guided Error Localization. Finally, we test whether structured, checklist-style queries can localize protocol deviations and maintain causal evidence across long horizons. Compared with a Qwen-based baseline that either produces generic descriptions or raises long-context memory errors during full-length processing, our parallel streaming architecture remains stable and continues to report evidence-grounded findings (Fig. 9). Across key decision points in medication administration, our system aligns responses to observable evidence (e.g., reviewing orders, verifying required checks) and retains the causal thread to later outcomes (e.g., subsequent abnormal bleeding consistent with missing verification), supporting actionable error localization in long clinical workflows.

Result 4: Live data collection for training and evaluation. In the January 2026 data collection session, standardized patients and care teams comprising an attending physician, two residents, and a nurse participated in six simulated scenarios (Syncope, Anaphylaxis, Pyelonephritis, etc.) at the SUNY Upstate medical care simulation center. Here are some details about one scenario that shows the high fidelity of actions performed at the center:

Syncope simulation: In the Simulation Center, the physician managed a standardized patient admitted after a witnessed syncopal episode, beginning with a stable bedside interview that intentionally includes nursing presence. During the encounter, the patient becomes lightheaded with position change and attempts to stand, creating a realistic fall-risk and rapid clinical deterioration sequence (near-syncope/collapse with hypotension and oxygen desaturation). The activity targets both clinical and teamwork behaviors: establishing rapport, obtaining a focused history of syncope/orthostatic symptoms, recognizing and mitigating fall risk before mobilization, calling for help early, assigning roles, repeating and interpreting vital signs, initiating immediate supportive measures (e.g., oxygen), and determining when to activate a rapid response. Physician is also expected to provide a succinct handoff to nursing and any responding team. (Figure 5.) These simulations validated the feasibility of synchronized multimodal capture, highlighted workflow and instrumentation challenges, and demonstrated that clinically meaningful sub-optimal behaviors can be reliably scripted and recorded.

5 Evaluation Plan

5.1 Technical Performance Evaluation.

The evaluation plan is designed to rigorously assess the proposed multimodal AI models and systems along accuracy, scalability, latency, robustness, and deployability dimensions, with emphasis on realistic hospital operating conditions. First, we will evaluate the correctness of multimodal understanding tasks, including action recognition, actor-action attribution, workflow-state identification, and protocol compliance checking. Metrics will include precision/recall and F1 for event

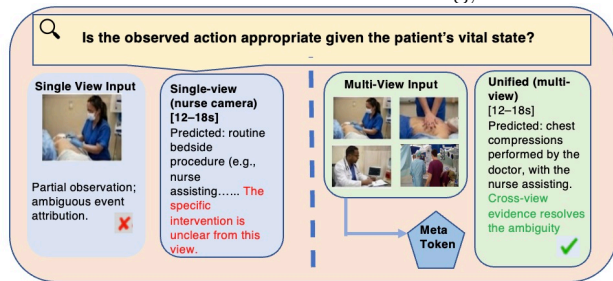


Figure 8: Multi-view fusion resolves actor-action ambiguity that is unresolved under single-view input.

Figure 8: Multi-view fusion resolves actor-action ambiguity that is unresolved under single-view input.

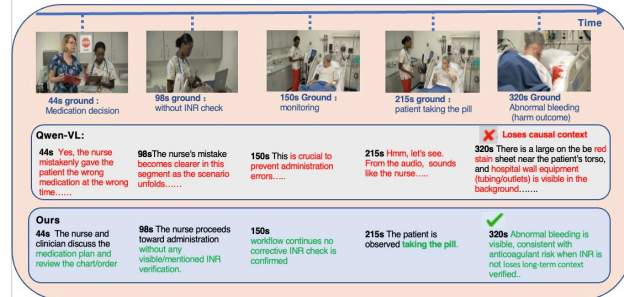


Figure 9: Query-guided error localization

detection, and accuracy for protocol adherence judgments. Second, to assess scalability to long-duration and multi-view settings, we will evaluate performance on continuous clinical encounters exceeding 30-60 minutes with multiple concurrent camera streams. Third, we will benchmark end-to-end system performance on target edge devices. Key metrics include end-to-end latency, alert response time (including tail latency), throughput (events/minute), memory footprint, and token/state growth rate over time. **Success criteria:** The system will be considered successful if it achieves high precision in event detection, protocol compliance checking from multi-view inputs, maintain stable performance over an extended time (8 hours+) on edge devices.

5.2 Real-world application testing

A central goal of this project is to evaluate system performance in live hospital settings using video and wearable sensor data collected during consenting care sessions. Evaluation will be conducted at two complementary levels. **Technical validity and agreement with the experts** The AI system will identify candidate inefficiencies and sub-optimal or failed care delivery events from live data. A panel of board-certified physicians will independently review and annotate all identified instances. System performance will be assessed using agreement with expert annotations (e.g., precision, recall, and inter-rater agreement). **Operational Relevance** Aggregated and de-identified findings will be reviewed with hospital administrators and clinical operations leaders. These stakeholders will assess whether the identified events are (i) actionable within existing workflows, (ii) feasible to address with reasonable effort, and (iii) likely to yield meaningful improvements in efficiency, safety, or patient outcomes. Evaluation at this level emphasizes prioritization and practical utility rather than raw detection volume. **Success criteria:** The system will be considered successful if it consistently surfaces a small set of high-confidence, expert-validated events that clinical and administrative stakeholders judge to be actionable and impactful. This evaluation explicitly prioritizes useful information over exhaustive detection, ensuring that the system supports real-world decision-making rather than generating excessive or low-value alerts.

6 Broader Impacts

Research Impact

This research advances the foundations of multimodal AI by enabling continuous, real-time understanding of complex hospital workflows through the unified modeling of video, audio, wearable sensing, and electronic health record data. It contributes new algorithms and representations for long-duration and multi-view reasoning, addressing key limitations in scalability, provenance preservation, and temporal consistency that constrain existing multimodal systems. The project further advances trustworthy computing by introducing privacy-preserving, cost-efficient edge deployment and online adaptation strategies that allow multimodal AI to operate under strict regulatory, latency, and resource constraints. By grounding these technical contributions in near-realistic simulations and live hospital evaluations, the research bridges the gap between laboratory benchmarks and operational environments.

Healthcare Impact: The three PIs will organize a dedicated AI + Healthcare session at a major medical conference to share project findings and foster dialogue among clinicians, hospital administrators, and AI researchers. Research outcomes will also be communicated directly to hospital leadership through focused briefings and round-table discussions, supporting actionable strategies to improve clinical workflow efficiency, patient safety, and patient outcomes.

Educational Impact Our educational activities will encompass student training and curriculum development, emphasizing the involvement of underrepresented minority groups.

Graduate Student Training: We anticipate that this project will directly support two graduate research assistants. In addition to regular graduate training in each PI's technical area, we will also

provide training for their career development to prepare them for future academic or industrial careers. The CS students and medical trainees will also get an opportunity to collaborate to broaden their research perspectives. Student participants also have opportunities to study multimodal AI, programming, and performance optimization on edge devices. This project will establish collaborations for students at the University of Houston (UH) and SUNY Upstate Medical University.

Undergraduate and K-12 Student Training: The PIs will actively recruit and engage undergraduate and K-12 students, ensuring a supportive environment for all participants. PI Gnawali is the *coordinator for the undergraduate research program and events* at the department and will bring research opportunity from this project to the students. He is supervising one Bachelors thesis and mentoring three undergraduate research projects related to AI system optimization. PI Zhang consistently invites undergraduate students to his research labs and offers them research opportunities in machine learning algorithms and computer systems. PI Zhang advises two undergraduate students on enhancing the interaction capabilities of MLLMs with streaming video content. Both PIs will work with the college and university undergraduate initiatives to recruit students and to disseminate the research outcomes to the broader undergraduate student community (UH Research day) and department events (Research Showcase). In terms of K-12 education, PI Zhang now actively guides a junior student at William P. Clements High School in developing an MLLM to track human movement. PI Zhang will also give hands-on Introduction to Machine Learning courses, covering LLMs and MLLMs, to students from William P. Clements High School. PI Zhang will continue collaborating with the university’s research events and summer camps program to engage more students interested in machine learning and computer systems. PI Gnawali has in the past spoken at high school, mentored high school research students, and will continue such activities.

Curriculum Development: This research will enhance the teaching of undergraduate and graduate courses related to computer system, machine learning, parallel and distributed computing, and edge computing. Specifically, PI Omprakash will integrate materials related to edge computing and edge networks into his undergraduate course, COSC 4377 *Introduction To Computer Networks*. PI Zhang will incorporate materials related to computer system and machine learning into his COSC 6342 *Machine Learning* and COSC 6374 *Parallel Computations* courses at UH. Additionally, PI Omprakash will design a *Special Topics Course on Edge Computing*. Course projects will be designed around efficient edge programming and communication. PI Zhang will establish a new course on AI accelerator, COSC 6436 *Efficient Machine Learning System*. PI Lamichhane will help curate representative use cases, co-develop instructional modules (lectures, labs, and project-based assignments), and participate in guest lectures and mentoring to support interdisciplinary student/resident/fellows learning. It will help improve simulation and real-world-based teaching and learning activities at the University Hospital. These contributions will enable rigorous, application-driven educational experiences that prepare learners to design, evaluate, and deploy trustworthy solutions at the edge in real operational environments.

Outreach Activities The PIs will organize a series of outreach activities as follows.

Training Program: The PIs have collaborated with multiple industries, government research labs, and medical universities, such as Microsoft Research, Meta Research, Argonne National Laboratory, Pacific Northwest National Laboratory, and SUNY Upstate Medical University. They are in an ideal position to organize a training program for scientists, residents, and healthcare providers from both universities and national labs. Participants will be trained using resulting insights, experiences, and software products (e.g., MLLMs fine-tuning and deployment, multi-view supporting, edge computing) from this project, helping scientists, residents, and healthcare providers learn how to fine-tune and deploy MLLMs in this project and apply them to advance critical applications. The PIs plan to initially launch this training program among graduate students, residents, and healthcare providers at UH and SUNY Upstate and then gradually promote it nationwide.

Outcome Dissemination: The PIs will actively engage with the public and share their research findings through various channels. First, the PIs will participate in conferences, workshops, and regional science fairs to present their research findings and engage with the public. Second, both PIs have given numerous invited talks and tutorials at conferences and workshops. In the next three years, the PIs will organize a set of workshops, special sessions, and one-day tutorials at major conferences, such as (1) efficient MLLM for multi-view and stream video at NeurIPS, CVPR, ICLR, and ICML. (2) Efficient MLLM fine-tuning and deployment on edge devices at MLsys, SC, PPOPP, IPDPS, DAC. They plan to attract speakers and audiences from academia, national labs, and industry to broaden the technical impact of their work. Third, they will maintain a project website where they will publish research papers, technical reports, presentations, and software tools developed during the project.

Table 1: Activity timeline. H indicates effort by two UH students; S indicates effort by one SUNY Upstate student.

Task	Task Description	Timeline (Year)					
		0.5	1	1.5	2	2.5	3
1	Thrust 1: Advancing Hospital-Floor Customized Medical Multimodal AI Model						
1.1	Fine-tune model to understand hospital context.	H	H				
2	Thrust 2: Advancing Support for Multiple Camera Views and Long Stream Video						
2.1	Support multiple camera inputs.		H	H			
2.2	Support long-stream video.			H	H		
3	Thrust 3: Advancing the System for Hospital Data Privacy and Cost						
3.1	Edge-based deployment of multimodal models.			H	H		
3.2	Online and continuous fine-tuning on edge.				H	H	
4	Thrust 4: Real and Live Data Collection and Evaluation						
4.1	Medical data collection.	S	S				
4.2	Address lack of near-realistic and imbalanced multi-modal data.	H/S	H/S				
4.3	Real-world evaluation.					H/S	H/S

7 Project Timeline

The team will collaborate closely and maintain frequent communication through online professional meetings and campus visits. PI Gnawali will coordinate meetings that require attendance from both UH and Upstate sites, ensuring efficient project progress management.

Table 1 presents the activity timeline. In the first year, UH will focus on fine-tune model to understand the hospital context via alignment across multiple modalities. SUNY Upstate will focus on medical data collection. UH will also make the model support for multiple camera views and long stream video. During the second year, we will establish efficient methods for deploying the multimodal model onto edge devices to achieve privacy-preserving, cost-efficient, and real-time inference, and continue updating. Finally, in the third year, we will evaluate the effectiveness of the proposed solutions (i.e., performance and accuracy) on realistic hospital workflows.

8 Results from Prior NSF Support

Omprakash Gnawali (PI): Two awards *NSF Student Travel Grant for 2020 Cyber-Physical Systems and Internet-of-Things Week (CPS-IoT Week)* and *Conference: CPS: NSF Student Travel Grant for 2024 Cyber-Physical Systems and Internet-of-Things Week (CPS-IoT Week)* enabled 22 and 13 students respectively from the US institutions to attend the CPS-IOTWEEK conference. The event provided the participants an opportunity to learn about the field, present their research, and professional development.

Chengming Zhang (PI) is a junior faculty member and has no prior NSF support.

Jivan Lamichhane (PI): No prior NSF support.

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